

ENBC332: Homework #8

Due: Monday, 11/20/25 by the end of the day

Submit your homework to ELMS, including R code and dataset

Overall points: 50

Note: Late homework may be submitted up to 24 hours after the due date, but a 50% penalty will be applied.

Note: Please aim to write clean, well-organized code that is easy for others to understand. Be sure to include comments or explanations where needed. Points may be deducted if this is not done.

Note: Please name your codes using the following format: "problem1_fullname.R". Start with the problem number and followed by your full name, e.g., "**problem1_jimmyazarnoosh**".

Note: Using generative AI for learning purposes is allowed. However, directly copying or closely replicating AI-generated content is strictly prohibited and will result in a grade of **ZERO**. Repeated violations may lead to more severe consequences.

Note: Submit **a single zip file** containing the following:

- The dataset you used
- R code files
- Your answers to the questions in PDF format

Note: Functions are recommended to follow the format below:

e.g., calculating range.

```
# -----  
Data <- trees$Height  
my_range_func <- function(Data){  
  Range <- max(Data) - min(Data)  
  return(Range)  
}  
# -----
```

Simple Linear Regression

Problem 1. You are asked to analyze data collected from a small bakery to understand the relationship between the number of advertisements they run each month (x , in units) and their monthly sales (y , in \$1000). (30 points)

The bakery provides the following data for the last 6 months:

Month	Number of Advertisements (x)	Monthly Sales (y) in \$1000
1	1	3
2	2	5
3	3	7
4	4	10
5	5	12
6	6	14

Do the following tasks:

a) **Scatter Plot:** Plot the data points on a graph with x on the horizontal axis and y on the vertical axis. Discuss the trend you observe. (5 points)

b) **Fit the Model:** Compute the slope (b_1) and intercept (b_0) of the regression line using the formulas and write the equation of the regression line. (10 points)

- I. Write your own custom function to calculate the coefficients b_0 and b_1 .
- II. Use R function `lm()` to calculate coefficients.

Note: You should get similar results in both I and II.

c) **Prediction:** Use the regression equation to predict the sales if the bakery runs 4 and 8 advertisements in a month. (5 points)

d) **Calculate R^2 :** Compute the total sum of squares (SST), the regression sum of squares (SSR), and the residual sum of squares (SSE) to find R^2 . (10 points)

Note: Your calculated R^2 should match with R^2 reported in `summary()`.

Regression Diagnostics

Problem 2. A hospital is studying the relationship between the Body Mass Index (BMI) of patients (x) and their systolic blood pressure (SBP), (y) to assess potential risk factors for hypertension. Before confirming the results, the hospital's data team needs to ensure that the assumptions of linear regression are valid. (20 points)

The dataset is provided in the following:

```
# Given dataset
set.seed(123)
BMI <- rnorm(300, mean = 25, sd = 5)           # Predictor variable (BMI)
SBP <- 90 + 2.5 * BMI + rnorm(300, mean = 0, sd = 8) # Response variable (SBP)
```

Do the following tasks:

a) **Fit the Model:** Use the dataset provided to fit a linear regression model, i.e. `lm()`, predicting SBP from BMI. (5 points)

b) **Check assumptions** of (1) linearity, (2) normality of residuals, and (3) constant variance of errors. Explain your observation from these assumptions. If any of the assumptions are violated, suggest potential remedies (e.g., data transformation, inclusion of interaction terms, or a different model). (15 points)